

Food Remedy API
Research Report – RE007

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Enhancing User Understanding of Calorie Information in Food Applications

1. Introduction

Calorie information is commonly shown in food and health applications to help users make better dietary decisions. Apps such as MyFitnessPal, Uber Eats, and HealthifyMe present calorie values in different formats, including per serving, per 100g, and daily tracking systems. However, although this information is available, many users still find it difficult to understand because calorie numbers are often shown without enough context. This report examines how calorie data is presented, the challenges users face in interpreting it, and how the Food Remedy UI can improve calorie understanding.

2. Presentation of Calorie Information in Existing Applications

Food applications usually present calorie data in three main ways.

Per serving:

This format is common in tracking apps such as MyFitnessPal. It is easy to read and practical for meal logging, but users may not fully understand what one serving means, especially when portion sizes differ.

Per 100g / 100ml:

This method is common on packaged foods and in some nutrition apps. It allows accurate comparison between products because it is standardized. However, it is less intuitive for everyday users and often requires mental calculation.

Per item or dish:

Food delivery apps such as Uber Eats often show calories for a full item, such as a pizza or burger. This is simple and quick to understand, but it may not reflect shared portions or ingredient breakdowns.

Some applications also use **daily calorie tracking**, showing calories consumed and calories remaining. This gives users more context, but it can be overwhelming for beginners.

3. User Difficulties in Understanding Calorie Data

One major issue is the **lack of context**. A value such as 300 kcal may not clearly tell the user whether the food is a low, moderate, or high-calorie choice.

Another problem is the **misunderstanding of serving sizes**. Users may assume a serving matches the amount they actually eat, which can lead to inaccurate calorie tracking.

Users may also experience **cognitive overload** when too many numbers are shown together, such as calories, fats, carbohydrates, and sugar. This can make decision-making harder, especially for new users.



Figure: Example of complex nutrition labels that can confuse users due to serving sizes and multiple values.

A further challenge is the **lack of personalisation**. Calorie needs differ depending on age, activity level, gender, and health goals. Generic calorie information may therefore feel irrelevant.

Finally, users may **misinterpret calorie values**, for example assuming that low-calorie food is always healthy or that high-calorie food is always unhealthy. This ignores overall nutritional quality.

4. Methods to Improve Calorie Understanding

Calorie information can be made easier to understand through clearer and more visual presentation methods.

Visual indicators such as traffic-light colour coding can quickly show whether a food is low, moderate, or high in calories. This reduces the effort needed to interpret numbers.

Daily intake percentages can give context by showing how much a food contributes to a person's recommended daily intake. For example, 250 kcal can be displayed as 12% of daily intake.

Comparative information can make calorie values more relatable, such as showing that a food is equivalent to a short walk or another familiar food item.

Progress bars can help users see how much of their daily calorie allowance has already been used and how much remains.

Simplified labels and icons can further improve understanding by using terms such as “Low Calorie” or “Energy Dense” instead of relying only on numbers.

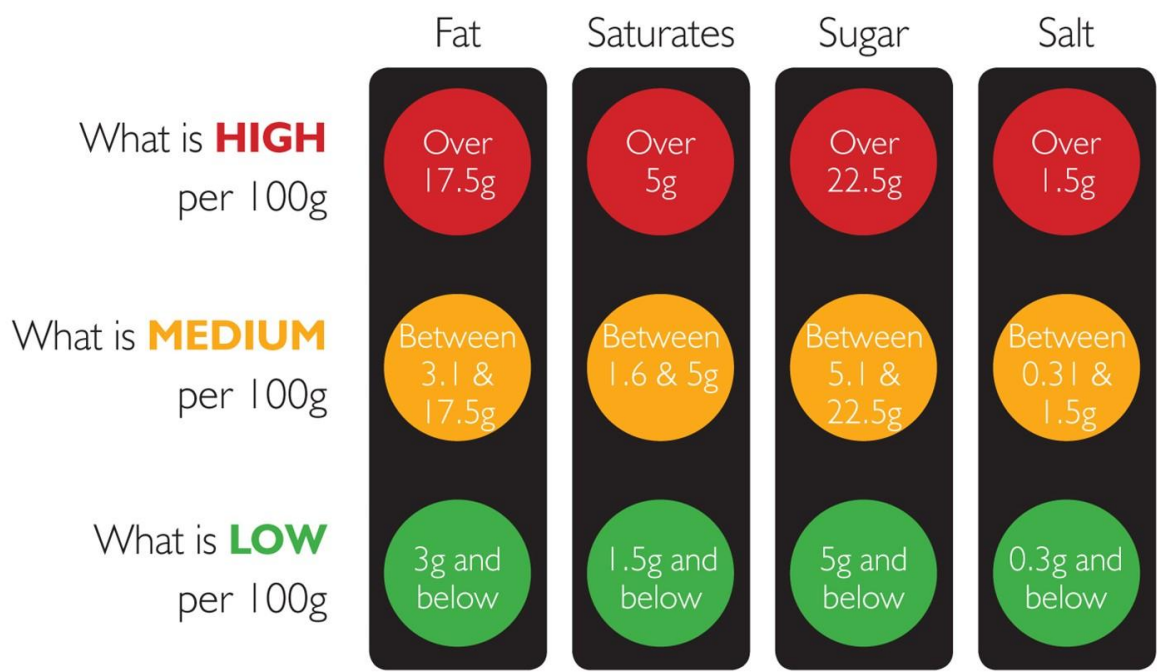
Personalised insights can improve relevance by adjusting calorie guidance according to a user’s goals, such as weight loss, maintenance, or fitness improvement.

5. Recommendations for Food Remedy UI

Based on this analysis, Food Remedy should adopt several design improvements to make calorie information easier to understand.

First, the app should use a **dual calorie display**, showing both per serving and per 100g values. This balances user friendliness and accuracy.

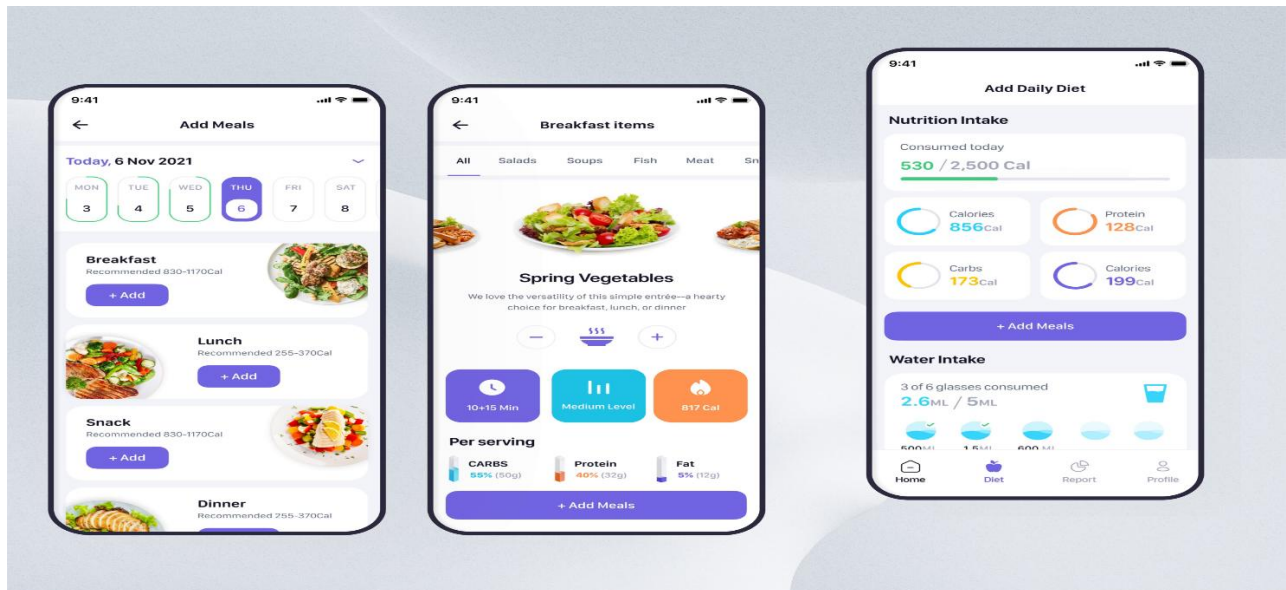
Second, a **traffic-light colour system** should be added so users can quickly identify whether a food is low, medium, or high in calories.



Based on guidelines by the Department of Health, under the terms of the Open Government Licence.

Figure: Improved calorie presentation methods including color coding, progress bars, and percentage indicators.

Third, the interface should display **daily intake percentages** so users can understand how a food fits into their total daily consumption.



Fourth, Food Remedy should include **smart comparisons**, such as simple activity or food equivalents, to make calorie values more meaningful.

Fifth, the UI should provide **interactive explanations**, such as tooltips or short labels, to help users understand serving sizes and calorie terms.

Finally, the app should support **personalised calorie insights** based on user preferences and health goals so that the information feels more relevant and useful.

6. Conclusion

Calorie information is most effective when users can easily understand and apply it. Although many food applications provide calorie data, the information is often shown in ways that are difficult to interpret. By using clearer formats, visual indicators, contextual comparisons, and personalized guidance, Food Remedy can improve calorie understanding and help users make healthier food choices.